Saintgits College of Engineering (Autonomous)

Biodata for internship in Taiwan under TEEP Scheme



1. Name of Student	Sidharth Sajith
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b. Email id (College)	sidharth.rbb2327@saintgits.org
c. Phone no	8089201760
d. Date of birth	28/04/2005
2. Course: B. Tech / M. Tech	B.Tech
3. Programme	Robotics and Automation
4. Batch	RB B
5. Current Semester	6
6. Semester at start of internship	7
7. Current CGPA	6.4

8. Career objectives	To leverage a strong foundation in robotics, edge computing, and AI to architect and deploy scalable, real-world hardware and software systems. Seeking roles that demand cross-functional expertise in computer vision, embedded systems, and IoT to drive innovation in autonomous navigation and intelligent automation.
9. Skill sets (in detail)	● **Robotics & Autonomous Systems:** Kinematics, autonomous navigation, sensor fusion, SLAM, behaviour tree logic, and sim-to-real transfer. ● **Artificial Intelligence & Machine Learning:** PyTorch, Scikit-Learn, Pandas, YOLOv8, YOLO26, Hugging Face (Transformers), OpenAI API, RAG (Retrieval-Augmented Generation) ● **Edge AI & Computer Vision:** Model quantization (TFLite), on-device inference (NPU/GPU), object detection, and integrating Large Language/Vision Models (LLMs/VLMs) on edge hardware (Jetson, Raspberry Pi). ● **Embedded Systems & IoT:** Microcontroller programming, real-time data streaming (MQTT), custom PCB design, hardware-software interfacing, and distributed sensor network architecture. ● **Full-Stack & Backend Integration:** Multi-threading, RESTful API integration, database management, and building real-time telemetry dashboards.
10. Proficiency in software tools	● **Programming Languages:** Python, C++, C, JavaScript (React.js). ● **Robotics & Simulation:** ROS 2 (Humble), NVIDIA Isaac Sim, Gazebo, Navigation2, URDF. ● **AI & Machine Learning:** TensorFlow, PyTorch, MediaPipe, OpenCV, YOLOv8, Google Gemini API, AWS Rekognition. ● **Hardware & Embedded:** PlatformIO, Arduino IDE, ESP-IDF. ● **Web & Database:** Flask, Streamlit, MySQL, MQTT Brokers.

11. Soft skills	● **Leadership & Team Management:** Proven track record of leading hardware teams under strict deadlines and managing university-level technical societies. ● **Rapid Prototyping:** Ability to quickly translate theoretical concepts into functional hardware/software prototypes. ● **Cross-Border Collaboration:** Experience working effectively in international, multi-stakeholder technical projects. ● **Complex Problem Solving:** Evidenced by debugging and optimizing autonomous workflows and simulation environments.
12. Experience (in detail)	● Serving as the Chairperson for The Robotics Society of India (RSI) - Saintgits Chapter, managing a team of student coordinators to bridge the gap between academic theory and industry practices. ● **Working as an AI & IoT Solutions Intern at a campus startup (Innovizta), developing multi-threaded AI desktop applications and full-stack IoT ecosystems (Smart Inhalers).** **Collaborating internationally as an Embedded Systems Research Intern for the ICCC Taiwan-India project, engineering distributed sensor networks and writing backend telemetry scripts.** **Served as a Core Executive Committee Member for SESA, coordinating departmental events and acting as the primary liaison between faculty and the student body.**
13.Online Certifications	● **AI Impact Summit Buildathon** – HCL GUVI (Mar 2026) ● **Introduction to Internet of Things** – NPTEL ● **Introduction to Solid Edge** – Siemens (Issued Oct 2024) ● **Fundamentals of Python Programming** – CODE@Saintgits (Issued Dec 2023) ● **Prompt Design in Vertex AI – Google Cloud (Jun 2025)** ● **Advanced Ethical Hacking and Cyber Defense – TECHMAGHI (Jul 2024)***)* ● **Getting Started with AI on Jetson Nano** – NVIDIA (May 2025)

14.Major Achievements	● **National Top 50, e-Yantra Robotics Competition:** Successfully progressed through rigorous technical evaluations to achieve a Top 50 national ranking out of thousands of teams. ● **Hardware & Circuit Lead, Srishti 2025:** Led a 5-member team in a 4-day rapid prototyping hackathon, successfully delivering multiple functional prototypes (NeuraHome and Smart Fire Alarm) under strict time constraints.
15.Extracurricular activities	● Chairperson of The Robotics Society of India (RSI) - SAINTGITS Chapter. ● Active Volunteer of Baja SAE (A-Baja), contributing to vehicle system simulations and autonomous model integration. ● Former Executive Committee Member of SESA (Saintgits Engineering Students Association).
16.Internships done in past	● I**CCC Saintgits (Taiwan-India Collaboration) | Embedded Systems Research Intern** • Engineered a scalable 6-node distributed sensor network using NodeMCU ESP32S and MQTT telemetry. • Integrated RESTful APIs (Open-Meteo, OGD India) and Python multi-threading to synchronize asynchronous virtual reference data with high-frequency physical sensor streams. • Implemented backend scripts to parse incoming payloads and automatically generate MySQL logs for historical analysis. ● **Innovizta (Campus Startup) | AI & IoT Solutions Intern** • Engineered **NeuralHealth**, a multi-threaded desktop app integrating MediaPipe and TensorFlow for emotion and posture tracking, synced with ESP32 sensors. • Developed **Breathe+**, a full-stack IoT smart inhaler with a Streamlit dashboard via MQTT and automated Telegram alerts. • Built an Intelligent Document Automation tool using YOLOv8 and Gemini API for 3-way supply chain matching.

17.Project work (in detail)	● **Autonomous Vehicle Research Platform (ADAS)** • Architecting a cost-effective ADAS system by repurposing a Samsung Galaxy S23 as the primary edge compute unit for autonomous navigation. • Optimizing and quantizing object detection models for NPU deployment via TFLite to reduce inference latency. • Developing synthetic training environments in NVIDIA Isaac Sim for sim-to-real transfer of driving policies. ● I**CCC Taiwan-India Distributed IoT Sensor Network** • Engineered a scalable 6-node distributed environmental sensor network utilizing NodeMCU ESP32S microcontrollers and low-latency MQTT telemetry. • Integrated RESTful APIs (Open-Meteo, OGD India) via Python multi-threading to seamlessly synchronize asynchronous virtual weather data with high-frequency physical sensor streams. • Developed automated backend data pipelines, MySQL logging scripts, and a centralized real-time web dashboard for cross-nation environmental monitoring. ● **EduBot: Interactive Educational Robot** • Engineered a socially interactive humanoid robot deployed on a lightweight DietPi OS, designed to engage primary school students. • Deployed a full-stack AI pipeline (LLM, Speech-to-Text, Text-to-Speech) directly onto a Raspberry Pi 5. • Integrated AWS Rekognition for computer vision-based student identification and managed profiles via a custom admin portal. • Utilized advanced prompt engineering to facilitate safe, age-appropriate bilingual interactions in English and Malayalam. ● **Aries Radar System** • Migrated legacy Tkinter code to a Full-Stack Web Architecture (Flask + HTML5/JS) for browser-based hand tracking via MediaPipe. Optimized real-time data streaming to achieve sub-50ms latency.

18.Name of the Taiwanese University to which the application is submitted to	Da-Yeh University
19.Name of the host professor in the Taiwanese University	Chin- Tsung Chan
DECLARATION:	I hereby declare that the above-furnished information is true to best of my knowledge and belief.

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Signature	
Date	20/05/2026